



estimated **one bloc at a time**, each on its **own sample**: per-bloc OLS VARX* / reduced-rank VECMX* ML

linked by trade-weighted foreign variables

Trade-weighted foreign (“star”) variables

$$x_{i,t}^* = \sum_{j \neq i} w_{ij} x_{j,t}, \quad w_{ii} = 0, \quad \sum_j w_{ij} = 1$$

built from **observed** partner data · **fixed** (as-of) weights

**WEAKLY
EXOGENOUS**
(tested per bloc)

COMMON WINDOW

stacked solve on the
overlapping span (ragged
edges truncated) — **no**
model-based completion

Stack via country link matrices

$$z_{i,t} = W_i x_t, \quad A_{i0} = (I_{k_i}, -\Lambda_{i0})$$

$$G_0 x_t = a + \sum_{\ell=1}^p G_\ell x_{t-\ell} + u_t \quad G_0 = \text{stack}_i(A_{i0} W_i)$$

Invert the stacking matrix G_0 once $\Rightarrow$ reduced-form Global VAR

$$x_t = G_0^{-1} a + \sum_{\ell=1}^p (G_0^{-1} G_\ell) x_{t-\ell} + G_0^{-1} u_t$$